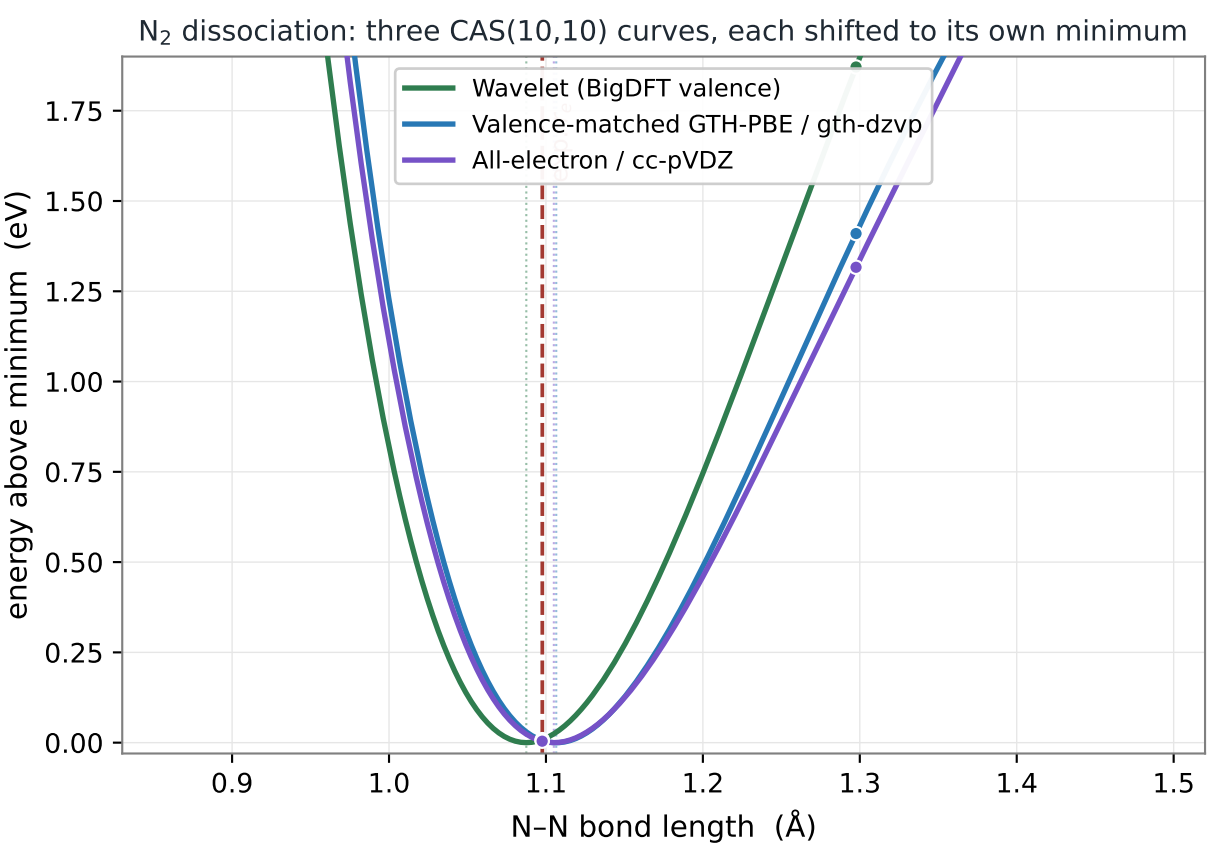

method	r_e (Å)	ω_e (cm^{-1})
Wavelet (BigDFT)	1.087	2549
GTH-PBE valence	1.107	2453
All-electron cc-pVDZ	1.105	2358
Experiment	**1.098**	**2359**

Zero-independent constants — comparable across the pseudopotential boundary (see note). r_e agrees to ~1%; ω_e is preliminary (coarse 0.2 Å scan, 5 virtuals). Wavelet from BigDFT integrals via the demo pipeline; references: PySCF CASCI(10,10); cc-pVDZ freezes two doubly occupied N 1s-derived molecular orbitals.